

CropSage Texas Crop Catalog

Twenty-two crop profiles for statewide Texas screening, presented as human-readable tables. All records are provisional pending Texas agronomist or Extension review.

Scoring guardrail: direct evidence may support deterministic scoring. Regional-transfer and generalized values are soft evidence only. A value marked not numerically established must never trigger a hard filter or numeric penalty.

Catalog Scope

Catalog version	Jurisdiction	Review date	Crops	References
1.0.0	Texas, United States	2026-08-26	22	29

Texas Regions

Region ID	Display name	Catalog interpretation	Sources
plains	Texas Plains	High Plains, Rolling Plains, and adjoining north-central production areas; generally semi-arid with important irrigated and dryland field-crop systems.	SRC-002, SRC-003
eastern	Eastern Texas and Gulf Coast	Humid eastern counties and the Gulf Coast production belt, including the principal Texas rice area and substantial forage and vegetable production.	SRC-002, SRC-003
central_south_winter_garden	Central and South Texas / Winter Garden	Central and southern production areas outside the Lower Rio Grande Valley, including the irrigated Winter Garden vegetable district.	SRC-002, SRC-003
lower_rio_grande_valley	Lower Rio Grande Valley	The subtropical southern tip of Texas, with mild winters, hot summers, and predominantly irrigated commercial agriculture.	SRC-002, SRC-003
far_west	Far West Texas	Arid Trans-Pecos and El Paso production areas where commercial cropping is strongly constrained by irrigation-water availability.	SRC-002, SRC-003

Region labels are screening zones, not legal boundaries or substitutes for county-specific Extension guidance.

The 22 Crop Profiles

One row represents one production-use profile. Planting windows are month-level screening guidance; county dates, cultivar maturity, soil temperature, irrigation, and forecast risk remain runtime checks.

Crop	Production use and scope	Texas regional suitability	Planting windows	Maturity	Temperature and frost	Soil	Water	Evidence
Upland cotton <i>Gossypium hirsutum</i> ID: upland_cotton	lint and cottonseed Commercial upland cotton cultivars used in Texas; Pima cotton is excluded.	Plains: primary Eastern: suitable Central/South: primary LRGV: primary Far West: conditional	Plains: May-Jun East: Apr-May C/S: Mar-May LRGV: Feb-Apr West: Apr-May	150-180 days Calendar-day planning range; final maturity is better tracked with 2,200-2,600 DD60 heat units.	Optimum: 26.7-32.2 C Heat: > 37.8 C (soft penalty) Duration: No defensible general-duration threshold was established for all Texas upland cultivars. Sensitive: flowering, boll set, boll development Frost: very high	Texture: loam, sandy loam, clay loam pH: 5.7-7 Drainage: well drained	Demand: high (700-1300 mm) Drought: high Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-005, SRC-006 Key assumption: Heat response varies by cultivar, humidity, soil moisture, and reproductive stage.
Corn grown for grain <i>Zea mays</i> ID: corn_grain	dry grain Commercial field-corn hybrids harvested for grain; sweet corn and silage endpoints are excluded.	Plains: primary Eastern: suitable Central/South: primary LRGV: suitable Far West: conditional	Plains: Apr-May East: Feb-Apr C/S: Feb-Apr LRGV: Jan-Mar West: Mar-Apr	110-140 days General commercial hybrid calendar range to grain maturity.	Optimum: 24-30 C Heat: > 35 C (soft penalty) Duration: Evaluate each hot day within the two weeks before and after tasseling; no universal consecutive-day cutoff is stored. Sensitive: tasseling, pollen shed, silking, early kernel set Frost: very high	Texture: loam, silt loam, clay loam, sandy loam pH: 5.5-7 Drainage: well drained	Demand: moderate (500-800 mm) Drought: low Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-005, SRC-007 Key assumption: The 35 C warning is a soft pollination-risk signal, not a statewide exclusion threshold.
Hard red winter wheat <i>Triticum aestivum</i> ID: hard_red_winter_wheat	food grain Texas-adapted hard red winter wheat cultivars; dual-purpose grazing effects are not modeled.	Plains: primary Eastern: suitable Central/South: suitable LRGV: limited Far West: conditional	Plains: Sep-Nov East: Oct-Nov C/S: Oct-Dec LRGV: Nov-Dec West: Sep-Nov	210-270 days Elapsed fall-to-summer calendar range, including winter dormancy.	Optimum: 23.9-25 C Heat: > 30 C (soft penalty) Duration: A few hot days near flowering can reduce yield, but the sources do not define one transferable numeric duration. Sensitive: five to seven days before flowering, anthesis, grain fill Frost: dormant hardy	Texture: loam, clay loam, silt loam, sandy loam pH: 5.5-7 Drainage: well drained	Demand: moderate (450-650 mm) Drought: moderate Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-005, SRC-008, SRC-009 Key assumption: Hard-red-winter class is represented at species level; cultivar-specific vernalization and disease resistance remain farm inputs.
Grain sorghum <i>Sorghum bicolor</i> ID: grain_sorghum	dry grain Commercial grain sorghum hybrids; forage and sorghum-sudangrass endpoints are separate records.	Plains: primary Eastern: suitable Central/South: primary LRGV: primary Far West: conditional	Plains: Apr-Jun East: Mar-May C/S: Feb-Jun LRGV: Feb-Mar, Jul-Aug West: Apr-May	90-120 days General commercial hybrid calendar range.	Optimum: 26-32 C Heat: > 37.2 C (soft penalty) Duration: Use heat events within roughly 15 days of flowering; no universal numeric duration is established. Sensitive: boot, flowering, early grain set Frost: very high	Texture: loam, clay loam, sandy loam, silty clay loam pH: 5.5-7 Drainage: well drained	Demand: moderate (450-650 mm) Drought: high Irrigation: often rainfed	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-005, SRC-009 Key assumption: Hybrid maturity and photoperiod response can shift the practical planting window.
Runner-type peanut <i>Arachis hypogaea</i> ID: runner_peanut	edible seed and oil Commercial runner market-type peanut cultivars; Virginia, Spanish, and Valencia market types are excluded.	Plains: primary Eastern: suitable Central/South: suitable LRGV: conditional Far West: limited	Plains: May-Jun East: Mar-May C/S: Apr-May LRGV: Mar-Apr	130-160 days Runner-type commercial maturity planning range.	Optimum: 25-30 C Heat: not established (informational only) Duration: Not numerically established; monitor combined heat and moisture stress during pegging and pod fill. Sensitive: flowering, pegging, pod set, pod fill Frost: very high	Texture: loamy sand, sandy loam, fine sandy loam pH: 5.8-7 Drainage: very well drained	Demand: moderate (500-700 mm) Drought: moderate Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-005, SRC-011 Key assumption: Runner market type is retained even though some Texas districts use other market types.
Long-grain rice <i>Oryza sativa</i> ID: long_grain_rice	milled long-grain rice Commercial long-grain lowland rice adapted to the US Gulf Coast; specialty and upland rice are excluded.	Plains: not supported Eastern: primary Central/South: limited LRGV: limited Far West: not supported	East: Mar-May	100-150 days General US Gulf Coast cultivar and management range.	Optimum: 25-30 C Heat: not established (informational only) Duration: Not numerically established; flowering and grain filling require stage-aware monitoring. Sensitive: panicle initiation, booting, flowering, grain fill Frost: very high	Texture: clay, clay loam, silty clay loam pH: 5.5-7 Drainage: poorly drained tolerant	Demand: very_high (450-700 mm) Drought: very low Irrigation: usually required	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-014 Key assumption: Seasonal water range is crop ET, not irrigation diversion or total field water use.
Soybean <i>Glycine max</i> ID: soybean	oilseed and protein meal Commercial soybean maturity groups selected for Texas latitude; food-grade specialty contracts are excluded.	Plains: conditional Eastern: primary Central/South: suitable LRGV: conditional Far West: limited	Plains: May-Jun East: Mar-Jun C/S: Mar-May LRGV: Feb-Apr	100-150 days Broad maturity-group planning range for Texas; cultivar days must replace this at recommendation time.	Optimum: 24-29.4 C Heat: > 29.4 C (soft penalty) Duration: The source says several subsequent days but does not establish a numeric count transferable to Texas. Sensitive: flowering, pod set, R5 seed fill Frost: very high	Texture: loam, silt loam, clay loam, sandy loam pH: 5.8-7 Drainage: well drained	Demand: moderate (450-700 mm) Drought: moderate Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-005, SRC-010 Key assumption: Maturity-group selection is a required local input before planting advice is actionable.
Grain oats <i>Avena sativa</i> ID: grain_oats	grain Fall-seeded grain oat cultivars adapted to Texas; forage-only oats are excluded.	Plains: suitable Eastern: primary Central/South: suitable LRGV: limited Far West: conditional	Plains: Sep-Oct East: Sep-Nov C/S: Oct-Nov	180-240 days Elapsed fall-to-summer calendar range, including cool-season slowdown.	Optimum: 20-21.1 C Heat: > 27.8 C (soft penalty) Duration: No transferable consecutive-day threshold was established; monitor hot periods before flowering and during grain fill. Sensitive: five to seven days before flowering, pollination, grain fill Frost: low	Texture: loam, clay loam, silt loam, sandy loam pH: 5.5-7 Drainage: well drained	Demand: moderate (450-650 mm) Drought: moderate Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-005, SRC-008 Key assumption: Record targets grain yield, not winter grazing or forage tonnage.
Oilseed sunflower <i>Helianthus annuus</i> ID: oilseed_sunflower	oilseed Commercial oilseed hybrids; confectionery sunflower is excluded.	Plains: primary Eastern: suitable Central/South: suitable LRGV: conditional Far West: conditional	Plains: Apr-Jul East: Mar-May C/S: Mar-Jun LRGV: Feb-Apr, Aug-Sep West: Apr-May	90-120 days General commercial oilseed-hybrid planning range.	Optimum: 15-25 C Heat: > 35 C (informational only) Duration: Not numerically established for whole-season oilseed production. Sensitive: germination, flowering, seed fill Frost: high	Texture: loam, silt loam, clay loam, sandy loam pH: 5.5-7.5 Drainage: well drained	Demand: high (600-1000 mm) Drought: high Irrigation: often rainfed	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-012 Key assumption: The temperature range is germination-focused and must not be treated as a full-season optimum-air band.
Sesame <i>Sesamum indicum</i> ID: sesame	oilseed and edible seed Non-shattering, mechanically harvestable commercial cultivars used in Texas and Oklahoma.	Plains: primary Eastern: conditional Central/South: primary LRGV: suitable Far West: conditional	Plains: May-Jun East: May-Jun C/S: Apr-Jun LRGV: Mar-May West: May-Jun	90-120 days General commercial non-shattering cultivar range.	Optimum: 25-32 C Heat: not established (informational only) Duration: Not numerically established. Sensitive: emergence, flowering, capsule fill Frost: very high	Texture: sandy loam, loam, silt loam pH: 5.5-8 Drainage: very well drained	Demand: low (not established) Drought: very high Irrigation: often rainfed	Confidence: low Status: provisional Sources: SRC-002, SRC-003, SRC-013 Key assumption: Numeric thermal and water thresholds remain a priority for Texas specialist review.
Corn grown for silage <i>Zea mays</i> ID: corn_silage	whole-plant silage Commercial silage hybrids harvested near whole-plant ensiling maturity; grain-harvest timing is excluded.	Plains: primary Eastern: suitable Central/South: suitable LRGV: conditional Far West: conditional	Plains: Apr-May East: Mar-Apr C/S: Feb-Apr LRGV: Jan-Mar West: Mar-Apr	80-110 days Planting-to-silage-harvest planning range; harvest is governed by whole-plant moisture and kernel stage.	Optimum: 24-30 C Heat: > 35 C (soft penalty) Duration: Evaluate hot days around tasseling and silking; no universal duration cutoff is stored. Sensitive: tasseling, silking, kernel set, early milk stage Frost: very high	Texture: loam, silt loam, clay loam, sandy loam pH: 5.5-7 Drainage: well drained	Demand: moderate (500-800 mm) Drought: low Irrigation: recommended	Confidence: medium Status: provisional Sources: SRC-001, SRC-003, SRC-004, SRC-005, SRC-007 Key assumption: Silage endpoint is whole-plant feed yield and moisture, not grain physiological maturity.
Forage sorghum <i>Sorghum bicolor</i> ID: forage_sorghum	hay, greenchop, or silage forage Commercial forage sorghum cultivars and hybrids; grain sorghum and sorghum-sudangrass are separate records.	Plains: primary Eastern: suitable Central/South: primary LRGV: suitable Far West: conditional	Plains: May-Jun East: Apr-Jul C/S: Mar-Jul LRGV: Feb-Aug West: May-Jun	60-90 days Planting to first forage harvest; regrowth and final silage maturity vary by hybrid.	Optimum: 26-32 C Heat: not established (informational only) Duration: Not numerically established for forage biomass production. Sensitive: emergence, rapid vegetative growth, boot stage for silage quality Frost: very high	Texture: loam, clay loam, sandy loam, silty clay loam pH: 5.5-7 Drainage: well drained	Demand: moderate (450-650 mm) Drought: high Irrigation: often rainfed	Confidence: medium Status: provisional Sources: SRC-003, SRC-004, SRC-005, SRC-009, SRC-026, SRC-028 Key assumption: Days to maturity means first usable harvest, not physiological grain maturity.
Sorghum-sudangrass <i>Sorghum bicolor</i> x <i>Sorghum bicolor</i> var. <i>sudanense</i> ID: sorghum_sudangrass	grazing, hay, greenchop, or silage forage Commercial sorghum-sudangrass hybrids; pure sudangrass and forage sorghum are excluded.	Plains: primary Eastern: suitable Central/South: primary LRGV: suitable Far West: conditional	Plains: May-Jun East: Apr-Jul C/S: Mar-Jul LRGV: Feb-Aug West: May-Jun	50-80 days Planting to first forage harvest; repeated-cut timing is management dependent.	Optimum: 26-32 C Heat: not established (informational only) Duration: Not numerically established for forage biomass. Sensitive: emergence, rapid vegetative growth, post-cut regrowth Frost: very high	Texture: loam, clay loam, sandy loam, silt loam pH: 5.5-7 Drainage: well drained	Demand: moderate (450-650 mm) Drought: high Irrigation: often rainfed	Confidence: medium Status: provisional Sources: SRC-003, SRC-004, SRC-005, SRC-009, SRC-026, SRC-028 Key assumption: Recommendation UI must warn that frost- or drought-stressed forage may require prussic-acid and nitrate management; toxicity is outside the suitability score.

Crop	Production use and scope	Texas regional suitability	Planting windows	Maturity	Temperature and frost	Soil	Water	Evidence
Alfalfa grown for hay <i>Medicago sativa</i> ID: alfalfa_hay	perennial hay Texas-adapted alfalfa cultivars managed for hay; grazing-only systems are excluded.	Plains: primary Eastern: limited Central/South: suitable LRGV: conditional Far West: primary	Plains: Sep-Oct, Mar-Apr C/S: Sep-Nov, Feb-Mar West: Sep-Oct, Feb-Mar	60-90 days Establishment to first hay harvest; established stands commonly reach bloom about 28-30 days after later cuttings.	Optimum: 18-30 C Heat: not established (informational only) Duration: Not numerically established. Sensitive: seedling establishment, post-cut regrowth, bud to early bloom Frost: dormant hardy	Texture: deep loam, silt loam, clay loam, well-drained sandy loam pH: 6.3-7.5 Drainage: very well drained	Demand: very_high (800-1600 mm) Drought: moderate Irrigation: usually required	Confidence: medium Status: provisional Sources: SRC-003, SRC-004, SRC-005, SRC-024, SRC-028 Key assumption: Days to maturity is first harvest after establishment, not stand lifespan.
Bermudagrass grown for hay <i>Cynodon dactylon</i> ID: bermudagrass_hay	perennial hay Seeded and vegetatively propagated forage bermudagrass cultivars; turf endpoints are excluded.	Plains: conditional Eastern: primary Central/South: primary LRGV: primary Far West: conditional	Plains: May-Jun East: Mar-Jun C/S: Mar-Jun LRGV: Feb-Jun West: Apr-Jun	60-90 days Establishment to first hay harvest under favorable conditions; established stands use cutting intervals rather than maturity days.	Optimum: 21.1-35 C Heat: not established (informational only) Duration: Not numerically established; heat risk is primarily mediated by water availability. Sensitive: new stand establishment, post-cut regrowth under drought Frost: moderate	Texture: sandy loam, loam, clay loam, silt loam pH: 5.7-7 Drainage: well drained	Demand: high (not established) Drought: very high Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-003, SRC-005, SRC-025, SRC-028 Key assumption: Variety cold tolerance and establishment method must be captured when a cultivar is chosen.
Annual ryegrass grown for forage <i>Lolium multiflorum</i> ID: annual_ryegrass_forage	cool-season grazing, baleage, or hay Annual Italian ryegrass cultivars used as winter forage; perennial ryegrass is excluded.	Plains: conditional Eastern: primary Central/South: suitable LRGV: conditional Far West: limited	Plains: Sep-Oct East: Oct-Nov C/S: Oct-Nov LRGV: Nov-Dec	60-90 days Planting to first grazing or cutting under adequate moisture; full season extends into spring.	Optimum: 10-21 C Heat: not established (informational only) Duration: Not numerically established. Sensitive: seedling establishment, spring vegetative growth, heading Frost: low	Texture: loam, silt loam, clay loam, sandy loam pH: 5.5-7 Drainage: moderate	Demand: moderate (not established) Drought: low Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-003, SRC-005, SRC-027, SRC-028 Key assumption: Days to maturity means first usable forage, not seed maturity.
White potato <i>Solanum tuberosum</i> ID: white_potato	fresh-market and processing tubers White-fleshed Irish potato cultivars; sweet potato is a separate record.	Plains: suitable Eastern: conditional Central/South: primary LRGV: suitable Far West: conditional	Plains: Mar-Apr, Jul-Aug East: Feb-Mar, Aug-Sep C/S: Jan-Mar, Aug-Sep LRGV: Nov-Feb West: Feb-Mar, Jul-Aug	90-120 days Texas variety and season planning range.	Optimum: 15-24 C Heat: not established (informational only) Duration: Not numerically established; sustained warm soil during tuber initiation and bulking is the primary concern. Sensitive: tuber initiation, tuber bulking, late-season skin set Frost: high	Texture: sandy loam, loam, fine sandy loam pH: 5-7.8 Drainage: very well drained	Demand: high (508-1016 mm) Drought: very low Irrigation: recommended	Confidence: medium Status: provisional Sources: SRC-002, SRC-003, SRC-015, SRC-020, SRC-023 Key assumption: The broad water range represents seasonal crop use, not a fixed irrigation prescription.
Sweet potato <i>Ipomoea batatas</i> ID: sweet_potato	fresh-market and processing storage roots Commercial sweet-potato cultivars established from slips; ornamental types are excluded.	Plains: conditional Eastern: primary Central/South: suitable LRGV: suitable Far West: limited	Plains: May-Jun East: Apr-Jun C/S: Mar-May LRGV: Feb-May	90-150 days Broad Texas cultivar and season planning range.	Optimum: 24-32 C Heat: not established (informational only) Duration: Not numerically established. Sensitive: slip establishment, storage-root initiation, storage-root bulking Frost: very high	Texture: fine sandy loam, sandy loam, loam pH: 5-7.5 Drainage: very well drained	Demand: moderate (254-508 mm) Drought: moderate Irrigation: conditional	Confidence: medium Status: provisional Sources: SRC-002, SRC-003, SRC-015, SRC-019, SRC-023 Key assumption: Days to maturity begins at slip transplanting, not mother-root bedding.
Seedless watermelon <i>Citrullus lanatus</i> ID: seedless_watermelon	fresh-market fruit Commercial triploid seedless hybrids established as transplants with a compatible diploid pollenizer.	Plains: suitable Eastern: suitable Central/South: primary LRGV: primary Far West: conditional	Plains: May-Jun East: Mar-Jun C/S: Feb-May LRGV: Jan-Apr West: Apr-May	65-75 days Days after transplanting in the Texas commercial crop guide.	Optimum: 26.7-35 C Heat: not established (informational only) Duration: Not numerically established; monitor heat with pollination and water status. Sensitive: transplant establishment, flowering and pollination, fruit set, fruit enlargement Frost: very high	Texture: loamy sand, sandy loam, light-textured loam pH: 5.5-8 Drainage: very well drained	Demand: moderate (254-381 mm) Drought: low Irrigation: recommended	Confidence: high Status: provisional Sources: SRC-002, SRC-003, SRC-015, SRC-016, SRC-023 Key assumption: Pollenizer compatibility and bee activity are required operational checks outside the climate-soil score.
Dry-bulb onion <i>Allium cepa</i> ID: dry_bulb_onion	cured dry bulbs Texas commercial dry-bulb onions; short-, intermediate-, or long-day cultivar class must match latitude.	Plains: suitable Eastern: conditional Central/South: primary LRGV: primary Far West: suitable	Plains: Feb-Mar East: Sep-Nov, Jan-Feb C/S: Sep-Nov LRGV: Sep-Nov West: Jan-Mar	150-210 days Seed-to-mature-bulb planning range; transplanting shortens field duration.	Optimum: 15.6-26.7 C Heat: > 35 C (informational only) Duration: Not numerically established for crop injury; establishment guidance is based on seed-zone temperature. Sensitive: seed germination, seedling establishment, bulb initiation, bulb enlargement Frost: moderate	Texture: sandy loam, loam, silt loam pH: 6-8.4 Drainage: very well drained	Demand: high (635-762 mm) Drought: very low Irrigation: usually required	Confidence: high Status: provisional Sources: SRC-002, SRC-003, SRC-015, SRC-017, SRC-023 Key assumption: A day-length cultivar class must be selected before regional suitability can become actionable.
Fresh-market cabbage <i>Brassica oleracea</i> var. <i>capitata</i> ID: fresh_market_cabbage	fresh-market heads Green fresh-market heading cabbage; processing, red, and specialty savoy classes are excluded.	Plains: suitable Eastern: suitable Central/South: primary LRGV: primary Far West: conditional	Plains: Feb-Mar, Jul-Aug East: Jan-Mar, Aug-Oct C/S: Sep-Nov, Jan-Feb LRGV: Sep-Dec West: Feb-Mar, Aug-Sep	70-120 days Transplant-to-harvest range varies strongly by cultivar and season.	Optimum: 15.6-21.1 C Heat: not established (informational only) Duration: Not numerically established. Sensitive: transplant establishment, head initiation, head filling Frost: low	Texture: loam, silt loam, clay loam, sandy loam pH: 6-7.5 Drainage: well drained	Demand: high (508-762 mm) Drought: very low Irrigation: recommended	Confidence: high Status: provisional Sources: SRC-002, SRC-003, SRC-015, SRC-021, SRC-022, SRC-023 Key assumption: Days to maturity is measured from transplanting where transplants are used.
Fresh-market spinach <i>Spinacia oleracea</i> ID: fresh_market_spinach	fresh-market leaves Commercial fresh-market spinach; processing and baby-leaf contract specifications are excluded.	Plains: suitable Eastern: conditional Central/South: primary LRGV: primary Far West: suitable	Plains: Feb-Mar, Aug-Oct East: Jan-Mar, Sep-Nov C/S: Sep-Feb LRGV: Oct-Feb West: Feb-Mar, Sep-Oct	35-50 days Fresh-market harvest planning range; cultivar and leaf-size target alter duration.	Optimum: 18.3-23.9 C Heat: > 23.9 C (informational only) Duration: Not numerically established; high seed-zone temperature is assessed at planting and sustained warm weather increases bolting risk. Sensitive: germination, stand establishment, leaf expansion, pre-bolting vegetative growth Frost: low	Texture: clay loam, loam, silt loam pH: 6.5-8 Drainage: well drained	Demand: moderate (254-381 mm) Drought: very low Irrigation: recommended	Confidence: high Status: provisional Sources: SRC-002, SRC-003, SRC-015, SRC-018, SRC-023 Key assumption: The catalog targets full-size fresh-market leaves rather than baby-leaf harvest.

Reference Registry

Reference IDs in the crop table resolve here. URLs are included for auditability and should be rechecked when a catalog record is reviewed or promoted to approved.

ID	Source	Supports	URL
SRC-001	Field Crops: Usual Planting and Harvesting Dates USDA National Agricultural Statistics Service government	Texas statewide field-crop planting windows; harvest timing	https://www.nass.usda.gov/Publications/Todays_Reports/reports/fcdte10.pdf Accessed 2026-08-26
SRC-002	Descriptions of Geographic Regions in Texas Texas A&M AgriLife Extension extension	Texas region vocabulary; regional climate and production context	https://aggie-horticulture.tamu.edu/vegetable/guides/the-crops-of-texas/appendix-1-descriptions-of-geographic-regions-in-texas/ Accessed 2026-08-26
SRC-003	The Crops of Texas: Highlights and Overviews Texas A&M AgriLife Extension extension	regional crop distribution; Texas production context	https://aggie-horticulture.tamu.edu/vegetable/guides/the-crops-of-texas/highlights-and-overviews/ Accessed 2026-08-26
SRC-004	Crop Water Needs and Sensitivity to Drought Food and Agriculture Organization of the United Nations international guidance	indicative seasonal crop-water ranges; relative drought sensitivity; critical growth stages	https://www.fao.org/4/S2022E/s2022e07.htm Accessed 2026-08-26
SRC-005	OSU Soil Test Interpretations Oklahoma State University Extension extension	Southern Plains crop soil-pH preferences	https://extension.okstate.edu/fact-sheets/osu-soil-test-interpretations Accessed 2026-08-26
SRC-006	When Is the Best Time to Plant Cotton in the Oklahoma Panhandle? Oklahoma State University Extension extension	cotton thermal basis; cotton heat warning; cotton maturity heat units	https://extension.okstate.edu/fact-sheets/when-is-the-best-time-to-plant-cotton-in-the-oklahoma-panhandle Accessed 2026-08-26
SRC-007	Dry Conditions During Corn Pollination University of Minnesota Extension extension	corn heat thresholds; pollination sensitivity; interaction between heat and soil moisture	https://extension.umn.edu/agriculture/crop-production/corn/dry-conditions-during-corn-pollination Accessed 2026-08-26
SRC-008	Heat Stress and Drought Stress in Small Grains University of Minnesota Extension extension	wheat and oat temperature ranges; small-grain heat-sensitive stages; small-grain heat thresholds	https://extension.umn.edu/growing-small-grains/heat-stress-and-drought-stress-small-grains Accessed 2026-08-26
SRC-009	Agriculture, Climate, and Change: Temperature Effects on Wheat and Sorghum Kansas State University Extension extension	wheat and sorghum thermal response; flowering heat sensitivity	https://eupdate.agronomy.ksu.edu/article/new-eupdate-series-on-agriculture-climate-and-change-part-1-298 Accessed 2026-08-26
SRC-010	How Do High Temperatures Impact My Soybeans? North Carolina State Extension extension	soybean optimum temperature; soybean heat stress; R5 sensitivity	https://soybeans.ces.ncsu.edu/news/how-do-high-temperatures-impact-my-soybeans/ Accessed 2026-08-26
SRC-011	Peanut Production Field Guide University of Georgia Cooperative Extension extension	runner peanut production requirements; soil and irrigation management; maturity	https://extension.uga.edu/publications/detail.html?number=B1146 Accessed 2026-08-26
SRC-012	Sunflower Planting Depth and Population in the Oklahoma Panhandle Oklahoma State University Extension extension	oilseed sunflower soil-temperature range; germination heat warning; drought tolerance	https://extension.okstate.edu/fact-sheets/sunflower-planting-depth-and-population-in-the-oklahoma-panhandle Accessed 2026-08-26
SRC-013	Sesame Seed Oil Properties Oklahoma State University Extension extension	commercial sesame scope; Texas and Oklahoma production context; drought tolerance	https://extension.okstate.edu/fact-sheets/sesame-seed-oil-properties Accessed 2026-08-26
SRC-014	Louisiana Rice Production Handbook LSU AgCenter extension	US Gulf Coast rice production requirements; soil, water, and crop calendar context	https://www.lsuagcenter.com/~media/system/9/0/e/9/0e93160aba5daccea90c6d955299f74/pub2331riceproductionhandbook2014completebook.pdf Accessed 2026-08-26
SRC-015	Appendix of Useful Tables: Organic Vegetable Production Guide Texas A&M AgriLife Extension extension	Texas vegetable water use; critical water stages; vegetable pH ranges; variety context	https://aggie-horticulture.tamu.edu/vegetable/guides/organic-vegetable-production-guide/appendix-of-useful-tables/ Accessed 2026-08-26
SRC-016	Seedless Watermelon Crop Guide Texas A&M AgriLife Extension extension	seedless watermelon temperature; soil; pH; water; maturity	https://aggie-horticulture.tamu.edu/vegetable/watermelon/seedless-watermelon-crop-guide/ Accessed 2026-08-26
SRC-017	Onion Commercial Crop Guide Texas A&M AgriLife Extension extension	dry-bulb onion temperature; day-length scope; soil; pH; planting conditions	https://aggie-horticulture.tamu.edu/vegetable/wp-content/uploads/sites/10/2011/10/onion1.pdf Accessed 2026-08-26
SRC-018	Spinach Commercial Crop Guide Texas A&M AgriLife Extension extension	fresh-market spinach temperature; soil; pH; frost tolerance	https://aggie-horticulture.tamu.edu/vegetable/wp-content/uploads/sites/10/2011/10/spinach.pdf Accessed 2026-08-26
SRC-019	Easy Gardening: Sweet Potatoes Texas A&M AgriLife Extension extension	sweet-potato temperature; soil; pH; frost response	https://aggie-horticulture.tamu.edu/vegetable/files/2010/10/EHT-026-Easy-Gardening-Sweet-Potatoes.pdf Accessed 2026-08-26
SRC-020	Easy Gardening: Irish Potatoes Texas A&M AgriLife Extension extension	white-potato planting temperature; Texas planting windows; soil and crop duration	https://aggie-horticulture.tamu.edu/wp-content/uploads/sites/12/2011/03/E-511_irish_potato.pdf Accessed 2026-08-26
SRC-021	Easy Gardening: Cole Crops Texas A&M AgriLife Extension extension	cabbage cold tolerance; Texas seasonal context	https://aggie-horticulture.tamu.edu/vegetable/wp-content/uploads/sites/10/2010/10/E-279_cole_crops.pdf Accessed 2026-08-26
SRC-022	Timely Tips on Starting Seedlings at Home Texas A&M AgriLife Extension extension	vegetable day and night growing temperature ranges	https://aggie-horticulture.tamu.edu/earthkind/landscape/starting-seedlings-at-home/ Accessed 2026-08-26
SRC-023	Fall Vegetable Gardening Guide Texas A&M AgriLife Extension extension	Texas vegetable planting zones; fall planting timing; frost susceptibility; maturity classes	https://aggie-horticulture.tamu.edu/wp-content/uploads/sites/10/2013/09/EHT-056.pdf Accessed 2026-08-26
SRC-024	Texas Alfalfa Production Texas A&M AgriLife Extension extension	alfalfa Texas adaptation; soil depth; water and drainage requirements; cutting interval	https://agrilifeextension.tamu.edu/wp-content/uploads/2026/03/texasalfalfaproduction.pdf Accessed 2026-08-26
SRC-025	Bermudagrass Varieties, Hybrids, and Blends for Texas Texas A&M AgriLife Extension extension	bermudagrass Texas adaptation; soil drainage; summer growth; hay use	https://agrilifeextension.tamu.edu/wp-content/uploads/2025/07/Bermudagrass-Varieties-and-Blends-for-Texas.pdf Accessed 2026-08-26

ID	Source	Supports	URL
SRC-026	Warm Season Annual Forages Texas A&M AgriLife Extension Forage Fax extension	forage sorghum and sorghum-sudangrass Texas adaptation; soil moisture and system fit	https://foragefax.tamu.edu/warm-season-annual-forages/ Accessed 2026-08-26
SRC-027	Cool Season Annual Forages: To Plant or Not To Plant Texas A&M AgriLife Extension Forage Fax extension	annual ryegrass Texas planting timing; soil-moisture requirement; forage season	https://foragefax.tamu.edu/cool-season-annual-forages-to-plant-or-not-to-plantnow-or-later/ Accessed 2026-08-26
SRC-028	Crop Evapotranspiration: Guidelines for Computing Crop Water Requirements Food and Agriculture Organization of the United Nations international guidance	crop coefficient context; water-demand comparison for field crops and forages	https://www.fao.org/4/x0490e/x0490e0e.htm Accessed 2026-08-26
SRC-029	SSURGO Web Soil Survey and Soil Data Access USDA Natural Resources Conservation Service government	runtime comparison of mapped soil texture; pH; drainage; available water capacity	https://www.nrcs.usda.gov/resources/data-and-reports/soil-survey-geographic-database-ssurgo Accessed 2026-08-26